

KRIPA SHANKAR

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PRINCIPAL AI ARCHITECT | ENTERPRISE AI ARCHITECT | PRINCIPAL AI ENGINEER | GENAI ARCHITECT

PROFESSIONAL SUMMARY

AI Architect with **15+ years of experience** designing and delivering **enterprise-scale AI platforms, cloud-native systems, and distributed architectures** across healthcare, automotive, BFSI and retail domains in software product R&D and technology consulting. Specialized in **Generative AI (LLMs, RAG, Agentic AI)**, multimodal AI systems, real-time data pipelines, and **production-grade AI architectures**. Proven ability to lead platform standardization, ensure governance and compliance (HIPAA, GDPR), and collaborate with global cross-functional teams to drive enterprise AI adoption.

Recognized for resolving process and technology bottlenecks, delivering business outcomes swiftly, exceptional performance and contributions in fast-paced workplaces, leading AI Coe initiatives driving accelerators and best practices.

ARCHITECTURAL EXPERTISE

- Designed and built **scalable, secured, observable AI-powered platforms** supporting multimodal workloads (text, voice, image)
- Foundation in **data integration, ETL/ELT processes**, and building batch and streaming pipelines using ADF, PySpark.
- Expertise in **distributed systems, microservices, and event-driven architectures**
- Built **production-grade GenAI modules** with RAG, fine-tuning, agentic workflows, 25+ production pull request in consecutive sprint.
- Strong experience in **cloud-native architecture (Azure, AWS, containers, CI/CD)**
- Drive Prototyping, POCs, and MVP** while working with **global teams and enterprise stakeholders for pre-sales support**

CORE TECHNOLOGY SKILLS

- AI & Machine Learning:** GenAI, LLMs, RAG, Agentic AI (AutoGen, CrewAI), NLP, Multimodal AI; Azure OpenAI, AWS Bedrock (Titan), Hugging Face, Mistral; TensorFlow, PyTorch, Keras, Scikit-learn; LangChain, LangGraph, Pandas, NumPy, Streamlit; Claude, Codex, GitHub Copilot, Hermes Agent, LLM Token Optimization, Agent Factory, Copilot Studio, Prompt Optimization
- AI Transcription & Evaluation:** Deepgram, Whisper AI, AWS HealthScribe, AWS A2I; Metrics: WER, CER, Latency, Accuracy, Confidence Scoring; Model benchmarking & optimization; Workflow Automation, MCP Architecture, A2A, AI Document Intelligence
- Cloud & Platform Engineering:** AWS, Azure, Containers, ECS, ECR, Lambda, S3, IAM; Microservices, Event-Driven Systems, Domain-Driven Design; CI/CD, Observability (logs, metrics, tracing); Secure architecture with PHI/PII redaction; Azure SQL DB, Cosmos DB, Kubernetes Service, Azure CLI & PowerShell, Azure SDK; Cloud-Native AI Solutions, Azure AI Foundry
- Data & Integration:** Azure Data Factory, Databricks, Synapse, PySpark; Real-time & Batch data pipelines; API design & integration.
- Programming:** Python, C#, .NET/.NET Core, SQL; JavaScript, ReactJS, FastAPI, Node.js; Semantic Search

PROFESSIONAL EXPERIENCE

Tao Digital Solutions – Technical Architect

Bengaluru | Aug 2025 – Present

- Architecting a **data-first multimodal AI platform** for transcription, summarization, and conversational intelligence.
- Designed enterprise AI platforms leveraging AI Agents, Multi-Agent Systems, LangGraph orchestration, Function Calling, OpenAI APIs, Model Context Protocol, AI Guardrails, AI Evaluation frameworks, and AI Observability to deliver secure, scalable, and production-ready healthcare AI solutions, IAM-based access control, LangGraph and OpenAI usage with deployment using AWS.
- Implemented **AI transcription pipelines** using Deepgram, Whisper AI, and AWS HealthScribe with **human-in-the-loop validation (AWS A2I)** for **Life Science and Healthcare** based research firm **Konovo**.
- Established **AI evaluation frameworks** (WER, CER, latency, accuracy), improving transcription quality by **40%+**.
- Built **60+ AWS Lambda** enabling scalable, event-driven workflows orchestrated with ECS, setting **end-to-end observability**
- Designed **sub-150ms low-latency distributed pipelines** using SQS and chunk-based processing.
- Implemented **secure, compliant AI architectures** with PHI/PII redaction with 8x to 10x speed of implementation with accuracy.
- Enabled reusable AI components and APIs for platform extensibility. Worked on identifying metrics for the model and setting evaluation and benchmarking to analyse using cloud services (Cognito, AWS IAM, Sagemaker, Cloudwatch, AWS S3, ECS).
- Engineered enterprise LLM platforms** with semantic caching, hierarchical chunking, context optimization, retrieval reranking, conversation memory management, and adaptive model selection which maximize model performance by 40% with scalability.
- Established enterprise AI governance frameworks** incorporating MCP-based tool interoperability, prompt lifecycle management, hallucination detection, benchmark evaluation, policy guardrails, auditability, telemetry, and continuous LLM quality monitoring.

- Architected enterprise LLMOps evaluation pipelines leveraging **vector embeddings**, automated **benchmark orchestration**, **comparative analysis** across Deepgram, Whisper, AWS HealthScribe, and OpenAI models, AI compliance validation, **workflow automation**, and **continuous quality monitoring (WER, CER, latency)** to accelerate model selection and deployment of production-ready transcription AI with optimal accuracy, scalability, and cost efficiency

Independent Consultant – AI Architect

Remote | March 2025 – Aug 2025

- Defined **enterprise AI platform strategies** and MVP roadmaps and designed **multi-agent AI systems** using AutoGen and CrewAI
- Delivered scalable AI-driven automation aligned with business outcomes.

Persistent Systems – Senior Architect

Bengaluru | Jul 2024 – Jan 2025

- Architected **GenAI-enabled enterprise platforms** integrating APIs, data pipelines, LLMs, and event-driven streaming using Kafka; built scalable microservices-based API platforms with Docker.
- Developed reusable **LLM accelerators and AI components**, ensuring data quality, governance, and system reliability.

EY – Senior Technical Lead / Architect

Gurgaon | Jul 2021 – Mar 2024

- Evaluated healthcare AI standards** by researching OpenEHR use cases for clinical research, decision support, and public health, guiding interoperable EHR architecture decisions.
- Led delivery and pre-sales initiatives** by driving estimation, process improvement, and mentoring teams across insurance, retail, and finance; supported **SOWs, RFPs, and POCs**, contributing to contract wins.
- Designed AI-driven architectures and POCs** using **Generative AI, NLP, chatbots, and speech-to-text**, accelerating client adoption and showcasing enterprise AI capabilities leveraging with python automation.
- Architected end-to-end cloud and AI platforms** on **Azure and AWS**, enabling real-time analytics, responsible AI implementation, observability, and scalable production deployments.
- Built and optimized GenAI solutions** leveraging **prompt engineering, fine-tuning, agentic AI, LLMs (SageMaker, Bedrock)**, reducing model training time by 40% and improving performance.
- Established production-grade MLOps and RAG pipelines** using Azure CLI, Cosmos DB, Azure SQL DB, **Azure ML, Databricks, ADF, Docker**, delivering multimodal RAG systems and compliant AI/ML lifecycle management.
- Delivered scalable data, API, and serverless solutions** with **ADF, Pandas/NumPy, Azure Functions, Azure APIM, .Net Core**, cutting preprocessing time by 40%, processing latency by 50%, and ensuring **GDPR/HIPAA** compliance.

Tata Technologies – Senior Technical Lead

Pune | Nov 2017 – Jun 2021

- Removed execution barriers using proactive **cloud adoption strategies**, achieving 0% downtime and 99% SLA ticket resolution.
- Led .NET team and **streamlined deployment processes**, delivering projects 30% faster.
- Built and **optimized CI/CD pipeline** with automated deployment and monitoring, reducing deployment times by 30%.
- Designed and presented AI solutions with pre-sales team, securing successful project bids and contract wins.

Genpact – Principal Consultant

Hyderabad / USA | Feb 2015 – Oct 2017

- Architected and led .NET solutions** to align with customer requirements and Agile practices, boosting user productivity by 60% and meeting SLA targets. Optimized and Improved post-deployment systems through analytics and cross-functional collaboration.
- Modernized legacy app** for 1,000+ users, achieving a 50% increase in productivity.
- Managed project delivery by task estimation, assignment, and mentoring, boosting team performance.

Annik Technology Services (Capgemini) – Lead Developer

Hyderabad | May 2012 - Jan 2015

North Shore Technologies – Programmer Analyst

Noida | Oct 2011 - April 2012

InvoSystems Solutions – Software Developer

Singapore | Nov 2009 - Dec 2010

AI INNOVATION HIGHLIGHTS

- Built AI personal assistant using AutoGen, streamlining **multi-agent task automation**, reducing manual task execution by 70%.
- Built **Enterprise Healthcare AI Platform** with SageMaker for healthcare analytics, optimizing resource planning and patient care.
- Developed **multimodal RAG** Platform with AWS **Bedrock** and FAISS, enhancing restaurant app search and recommendations.
- Created **Knowledge Graph RAG** with LlamaIndex and Python, delivering actionable financial market sentiment insights.
- Automated ticket processing with LLMs, Azure ML, Pandas, langchain-openai, Streamlit, and RAG, boosting customer efficiency

EDUCATION

- Master’s in systems and operations management, IIM Kozhikode (2012–2014)
- B.Tech. in Computer Science, GIET Gunupur (2002–2006)

CERTIFICATIONS

- Azure Solutions Architect Expert, Azure AI Engineer Associate, Azure Data Engineer Associate, Certified ScrumMaster (CSM)

PROJECT DETAILS

Org	Projects	Domain	Client	Assignments
Tao Digital Solutions	Incrowd Survey	Healthcare Research	Konovo	Agentic AI, AI Integrations, Model Evaluation and benchmarking, LLMOps, API Dev, POC, Production Deploy
Persistent	MDM Product, FM Backbone	Food Management	Sodexo	API Development, GenAI, POC
EY	Global Data Processing	BFSI	Global Atlantic	Data Engineering, Automation, Team Management, GenAI
	TMNAS	BFSI	Tokio Marine	Middleware, Team Management, Platform Architecture
	LYB Sustainability Certificate	ESG, Chemical	LyondellBasell	API Development, Data Engineering, GenAI, ML
	Sobeys	Retail	Sobeys	API Development, Data Engineering
	Q/RM	Auditing & Consulting	Internal	API Integration, Platform Architecture, POC
	GE Proficy-EV Battery	Manufacturing, IOT	Internal	IOT, Machine Learning, Cloud
	Retrofit	Entertainment	Seminole Hardrock	API Development, Monitoring
	6 POC			Prototype creation, GenAI, LLM, Python, Langchain
Tata Technologies	e-POD	Logistic, Manufacturing	Tata Steel	App Development, Cloud integration
	Comauzon	Automotive	FIAT	Cloud Migration, App Development, platform architecture, Project Management
	Innovista	Enterprise	Tata Sons	App Features Development
	PEOD	Automotive	Tenneco	App Features Development, Support, Project Management
Genpact	PEOD, GTR, WTR, WSO	Automotive	Tenneco	App Development, Support, platform architecture, Team Management, PM
Annik Inc	MDM, HCP	Product Development	Microsoft Corp.	App Development, Support, BI Integration
North Shore	Artcore Logistics	Logistics	Artcore Logistics	App Development
Invosystems	PSS-SAP Sync	Automotive	Astra International	App Development
Shah Infotech	Product Portfolio Application	Healthcare	Johnson & Johnson	Built frontend features